Hitesh H

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Education

Global Academy Of Technology — Bengaluru, IN
Bachelor of Engineering in Computer Science — *Sept. 2023 – July 2027*

- Cumulative GPA: 8.83/10.0

Projects

Hybrid Liver Tumor Detection Framework | *Python, PyTorch, ConvNeXt, U-Net, XGBoost* 2025 – 2026

- Developed a hybrid diagnostic framework using **2.5D ConvNeXt-Tiny** and **U-Net** for liver tumor detection and segmentation from CT scan images.
- Integrated clinical parameter analysis using **XGBoost** and implemented tumor localization through bounding boxes and segmented tumor regions.
- Applied **Grad-CAM** to generate heatmaps, providing visual explanations of the regions influencing model predictions.

MediAssist AI | *Python, FastAPI, React, Tesseract OCR, NLP* 2025 – 2026

- Developed a full-stack healthcare application for extracting and organizing information from medical prescriptions and documents using **OCR and NLP**.
- Implemented **Tesseract OCR** with image preprocessing to extract patient details, medications, doctor information, and prescription data.
- Built a **FastAPI backend** and React dashboard with REST APIs for structured data extraction, medical summaries, healthcare insights, and chatbot interactions.

REAGENT: Verified Document & Research Assistant | *Python, FastAPI, React, TypeScript, SQLite FTS5, LLM APIs*2026

- Developed a full-stack **multi-agent research and document Q&A system** with Archive for uploaded PDF/TXT/CSV files and Inquiry for web research through a plan, research, analyse, write, verify, and correct pipeline.
- Implemented **RAG using SQLite FTS5 and BM25 ranking**, enabling sentence-level source citations without embedding models or vector databases.
- Built a **claim-level verification and correction layer** that removes or rewrites unsupported claims, improving faithfulness from **0.92 to 1.0** in one evaluation run. Developed a streaming FastAPI backend with Gemini, Groq, and Anthropic fallback, evaluation tests, and a React + TypeScript frontend.

Activities

- Served as a **Web Development Team Member** for the **ACM Student Chapter**, contributing to responsive web projects and technical initiatives.
- Attended the **ROCS Research Outreach and Collaboration Series** at BMS College of Engineering and competed in ACM's **Prompt Battle** at GAT.
- Participated in an **Ethical Hacking and Cyber Security Workshop** organized by NIC and Global Academy of Technology (September 2023).

Technical Skills

Languages: Python, Java, SQL, HTML/CSS
Frameworks: FastAPI, React
Libraries: Pandas, NumPy, Matplotlib, Transformers, Scikit-learn
Developer Tools: Git, Docker, VS Code, Eclipse